瑕积分+参变量积分.

一. (1) $\int_1^2 \frac{\sqrt{x}}{\ln x}dx$

函数 $\frac{\sqrt{x}}{\ln x}$ 的瑕点为 $x=1$.

$\Rightarrow \lim\limits_{x\to1^+}(x-1)^p\frac{\sqrt{x}}{\ln x}=\lambda$

$\ln x \sim x-1$

取 $p=1$，则 $\lim\limits_{x\to1^+}(x-1)^p\frac{\sqrt{x}}{\ln x}=1$

$\Rightarrow$ 瑕积分收敛.

(2) $\int_0^1 \frac{1}{\sqrt{x}\ln x}dx$.

$\frac{1}{\sqrt{x}\ln x}$ 的瑕点为 0 和 1.

$\int_0^1\frac{1}{\sqrt{x}\ln x}dx=\int_0^{\frac12}\frac{1}{\sqrt{x}\ln x}dx+\int_{\frac12}^1\frac{1}{\sqrt{x}\ln x}dx$

第一个 $\int_0^{\frac12}\frac{1}{\sqrt{x}\ln x}dx$，瑕点为 0.

$x=0$，$\lim\limits_{x\to0^+}x^p\frac{1}{\sqrt{x}\ln x}$，无极限

$\int_{\frac12}^1\frac{1}{\sqrt{x}\ln x}dx$，瑕点为 1

$\lim\limits_{x\to1^+}(x-1)^p\frac{1}{\sqrt{x}\ln x}=\lim\limits_{x\to1^+}\frac{(x-1)^p}{\ln x}=\lambda=1$

$p=1$，$\lambda=1$. 瑕积分发散.

二. (1) $\lim\limits_{\alpha\to0}\int_{-1}^1\sqrt{x^2+\alpha^2}\,dx$

解 $\because \lim\limits_{\alpha\to0}\int_{-1}^1\sqrt{x^2+\alpha^2}\,dx=\int_{-1}^1\sqrt{x^2+0}\,dx=\int_{-1}^1|x|\,dx$

$=2\int_0^1x\,dx=2\cdot\frac12x^2\Big|_0^1=1.$

(2) $\lim\limits_{\alpha\to0}\int_0^2x^2\cos\alpha x\,dx$

$\lim\limits_{\alpha\to0}\int_0^2x^2\cos\alpha x\,dx=\int_0^2x^2\,dx=\frac13x^3\Big|_0^2=\frac83$

三. 求 $F'(x)$.

(1) $F(x)=\int_x^{x^2}e^{-xy^2}dy$

$F'(x)=-\int_x^{x^2}e^{-xy^2}y^2\,dy+e^{-x(x^2)^2}\cdot2x-e^{-x x^2}$

$=-\int_x^{x^2}e^{-xy^2}y^2\,dy+2xe^{-x^5}-e^{-x^3}$

(2) $F(x)=\int_{\sin x}^{\cos x}e^{x\sqrt{1-y^2}}dy$

$F'(x)=\int_{\sin x}^{\cos x}\sqrt{1-y^2}\,e^{x\sqrt{1-y^2}}dy+e^{x|\sin x|}(-\sin x)-e^{x|\cos x|}\cos x$

(3) $F(x)=\int_{a+x}^{b+x}\frac{\sin(xy)}{y}dy$

$F'(x)=\int_{a+x}^{b+x}\cos xy\,dy+\frac{\sin[x(b+x)]}{b+x}(b+x)'-\frac{\sin[x(a+x)]}{a+x}(a+x)'$

$=\left(\frac1x+\frac{1}{b+x}\right)\sin[x(x+b)]-\left(\frac1x+\frac{1}{a+x}\right)\sin[x(x+a)]$

(4) $F(y)=\int_0^1\ln(x^2+y^2)\,dx$

$F'(y)=\int_0^1[\ln(x^2+y^2)]'_{(y)}\,dx=\int_0^1\frac{2y}{x^2+y^2}dx$

$=2\arctan\frac1y$ $\quad\therefore F'(y)=2\arctan\frac1y.$

四. $\int_0^\pi \ln(1-2a\cos x+a^2)dx$. $|a|<1$

解: $I'(a)=\int_0^\pi \frac{2(a-\cos x)}{1-2a\cos x+a^2}dx$

$=\frac{\pi}{a}-\frac{1-a^2}{a}\int_0^\pi \frac{1}{a^2+1-2a\cos x}dx$

换元 $\tan\frac{x}{2}=t$

$\Rightarrow I'(a)=\frac{2}{1-a^2}\int_0^{+\infty}\frac{d(\frac{1+a}{1-a}t)}{(\frac{1+a}{1-a}t)^2+1}=\frac{\pi}{1-a^2}$

$\therefore I'(a)=\frac{\pi}{1-a^2}-\frac{\pi}{1-a^2}=0$

$\therefore I(a)=I(0)+\int_0^a 0\,du=I(0)=0$.

五. $\int_0^1 \sin(\ln\frac{1}{x})\frac{x^b-x^a}{\ln x}dx$. $a>0,b>0$

原式 $=\int_0^1\left[\sin(\ln\frac{1}{x})\int_a^b x^y\,dy\right]dx$

$=\int_a^b dy\int_0^1 x^y\sin(\ln\frac{1}{x})dx$

即对于 $I(y)=\int_0^1\sin(\ln\frac{1}{x})x^y dx$

$I(y)=\frac{1}{y+1}\int_0^1\sin(\ln\frac{1}{x})\,dx^{y+1}=\frac{1}{y+1}\sin(\ln\frac{1}{x})x^{y+1}\Big|_0^1-\int_0^1\cos(\ln\frac{1}{x})x^{y+1}(-\frac{1}{x})dx$

$=\frac{1}{y+1}\int_0^1\cos(\ln\frac{1}{x})x^y dx=\frac{1}{(y+1)^2}[1-I(y)]$

$\Rightarrow I(y)=\frac{1}{y^2+2y+2}$

原 $=\int_a^b\frac{1}{(1+y)^2+1}d(1+y)=\arctan(1+y)\Big|_a^b=\arctan\frac{b-a}{1+(1+a)(1+b)}$.

六 证明: $\int_0^1\frac{dx}{\sqrt{1-x^4}}\int_0^1\frac{x^2dx}{\sqrt{1-x^4}}=\frac{\pi}{4}$

令 $x^4=t$, $dx=d\sqrt[4]{t}=\frac{1}{4}dt^{-\frac{3}{4}}=\frac{1}{4}t^{-\frac{3}{4}}dt$.

$\int_0^1\frac{dx}{\sqrt{1-x^4}}=\frac{1}{4}\int_0^1 t^{-\frac{3}{4}}(1-t)^{-\frac{1}{2}}dt$

$=\frac{1}{4}B(\frac{1}{4},\frac{1}{2})$

$\int_0^1\frac{x^2dx}{\sqrt{1-x^4}}=\frac{1}{4}\int_0^1 t^{-\frac{1}{4}}(1-t)^{-\frac{1}{2}}dt$

$=\frac{1}{4}B(\frac{3}{4},\frac{1}{2})$.

$\therefore$ 原 $=\frac{1}{16}B(\frac{1}{4},\frac{1}{2})\cdot B(\frac{3}{4},\frac{1}{2})$

$=\frac{1}{16}\frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{\Gamma(\frac{3}{4})}\frac{\Gamma(\frac{3}{4})\Gamma(\frac{1}{2})}{\Gamma(\frac{5}{4})}$

$=\frac{1}{16}\frac{\Gamma^2(\frac{1}{2})\Gamma(\frac{1}{4})\Gamma(\frac{3}{4})}{\frac{1}{4}\Gamma(\frac{1}{4})\cdot\Gamma(\frac{3}{4})}$

$=\frac{1}{4}\Gamma^2(\frac{1}{2})=\frac{1}{4}\pi=\frac{\pi}{4}$.

$\therefore$ 原 $=\frac{\pi}{4}$